Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application

Aim

To analyze the Software Development Life Cycle (SDLC) models used in the development of a real-world software application and identify the most suitable SDLC model.

Objective

- Study different SDLC models.
- Analyze the development process of a real-world software application.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.

Selected Software Application

Library Management System:

Overview

A Library Management System is a software application that helps manage books, members, borrowing, returns, and fine collection. It enables students and librarians to perform library operations efficiently.

The system allows users to:

- Register/Login
- Search Books
- Borrow Books
- Return Books
- Pay Fine (if applicable)
- View Borrowing History

Functional Requirements

Student Module

- User Registration
- Login
- Search Books
- Borrow Books
- Return Books
- View Borrowing History

Librarian Module

- Add Books
- Update Book Details
- Issue Books
- Collect Returned Books

Admin Module

- User Management
- Library Reports
- Book Inventory

SDLC Phases

1. Requirement Analysis

- Student Registration
- Book Management
- Borrow & Return System

Deliverables

- Software Requirement Specification (SRS)

2. System Design

Design Includes

- Database Design
- User Interface Design

Deliverables

- UML Diagrams
- Database Schema

3. Development

Modules Developed:

- Login Module
- Book Management Module
- Borrow & Return Module

Programming Languages:

- Java/Python
- React
- Node.js

Database:

- MySQL / PostgreSQL

4. Testing

Testing Types:

- Integration Testing
- System Testing
- User Acceptance Testing (UAT)

5. Deployment

Deployment Options

- Cloud Deployment
- Mobile App Release
- Web Server Deployment

6. Maintenance

Activities

- Bug Fixes
- Security Updates
- Feature Enhancements

Conclusion

This case study shows that selecting the appropriate SDLC model depends on project requirements. For applications like a **Library Management System**, Agile provides flexibility, continuous improvement, and quick software delivery. DevOps complements Agile by automating deployment and maintenance. Together, Agile and DevOps improve software quality, user satisfaction, and long-term system reliability.